Alan Wang

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Research Interests

I have broad interests in side-channel attacks and defenses, AI for hardware security, computer architecture, security, and systems. In the past, I have also been interested in drug discovery and MD simulations.

Education

University of California, Berkeley Berkeley, California
Ph.D. in Computer Science 2025–2030

University of Illinois at Urbana-Champaign Urbana, Illinois
Conjoined BSMS in Computer Science 2021–2025

Publications

[1] Alan Wang et al. "Pixnapping: Bringing Pixel Stealing out of the Stone Age". In: *Proceedings of the 32nd ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 2025. URL: `https://www.pixnapping.com`.

[2] Siyuan Chai et al. "EMT: An OS Framework for New Memory Translation Architectures". In: *Proceedings of the USENIX Symposium on Operating Systems Design and Implementation (OSDI)*. 2025.

[3] Alan Wang et al. "Peek-a-Walk: Leaking Secrets via Page Walk Side Channels". In: *Proceedings of the IEEE Symposium on Security and Privacy (S&P)*. 2025.

[4] Rutvik Choudhary et al. "Declassiflow: A Static Analysis for Modeling Non-Speculative Knowledge to Relax Speculative Execution Security Measures". In: *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 2023.

[5] Alec Auster et al. "MITHRIL: Mars Ice Thermal Harvesting Rig and ISRU Laboratory". In: *Proceedings of AIAA ASCEND*. 2022.

Honors & Awards

- NDSEG Fellowship, *Department of Defense* 2026
- IEEE Micro Top Picks Honorable Mention, *IEEE* 2026
- Jane Street GRF Finalist, *Jane Street* 2026
- 3x High-Severity Bug Bounty Awards, *Google* 2025
- NSF GRFP Honorable Mention, *National Science Foundation* 2025
- Bronze Tablet Scholar Award, *University of Illinois at Urbana-Champaign* 2025
- Siebel Scholar Award, *Siebel Scholars Foundation* 2025
- Best Poster in Theme, *SRC PRISM* 2025
- Dean's List, *Grainger College of Engineering, UIUC* 2021–2023
- 2nd Place Overall, *NASA RASC-AL Competition* 2022

Bugs and News

CVEs: CVE-2025-48561, CVE-2025-48630

Highlighted news: Ars Technica, Malwarebytes, Risky Business, LWN

Experience

University of California, Berkeley — Berkeley, CA, USA
Graduate Student Researcher — August 2025 – Current

- Redacted: two works under submission
- Recent work: pixnapping.com. This works presents a new class of side-channel attacks on Android devices that can steal graphically rendered secrets. Most notably, we demonstrate leaking ephemeral app-based 2FA codes in under 30 seconds
- Attacks: researching side-channel attacks inherent to graphics stacks (i.e., pixel stealing attacks)
- Defenses: exploring defenses for the side-channel vulnerabilities found by my attack research

Jane Street — New York, NY, USA
Software Developer Intern — May – August 2024

- Worked with billions of trade orders submitted to the SEC
- Improved the efficiency and utilization of servers used by the entire firm

D. E. Shaw Research — New York, NY, USA
Computer Science Intern — May – August 2023

- Researched docking and non-equilibrium FE calculation methods
- Wrote system software which ran thousands of simulations on Anton3 ASICs and achieved a 2.5x performance improvement

FPSG Lab — Urbana, IL, USA
Research Assistant — September 2021 – May 2025

- Created a new class of side-channel attacks on Android devices.
- Created a new general-purpose cache side channel exploiting page walkers and used it to exploit the Linux kernel using Spectre-V2 and the Intel DDP
- Helped build a new Linux framework to support new, radical memory translation architectures
- Modelled non-speculative information flow to decrease mitigation overhead by up to 21%

Argonne National Lab — Lemont, IL, USA
Research Aide — May 2022 – May 2023

- Worked with NVIDIA's Bluefield-3 Data Processing Unit (DPU) for zero trust network architectures
- Found critical errors by instrumenting Portable Batch System (PBS) for Argonne's extreme scale systems
- Developed a command line interface for Argonne's UserBase3, used by all Argonne admins

Argonne National Lab — Lemont, IL, USA
Visiting Student — February – May 2022

- Led the design of a ROS2 interface for Argonne's self-driving lab
- Built key infrastructure for Argonne's self-driving lab

Northwestern University — Evanston, IL, USA
Undergraduate Research Assistant — February 2021 – June 2022

- Researched the vulnerability INTEL-SA-00086 to gain access to Intel's most secure piece of hardware (microcode project)

Argonne National Lab Lemont, IL, USA
DoE College Bound Research Intern (CBRP) June – August 2021

- Started the design of a ROS2 interface for Argonne's self-driving lab
- Created important building blocks for future work in Argonne's self-driving lab

Office of Naval Research Crane, IN, USA
SEAP Intern Summers 2019 & 2020

- Led the development of autonomous bomb-defusing robots
- Repaired software related to navigation, object recognition, and arm manipulation

Mentoring

- Jonathan Ji, *Undergrad Research Mentor* 2026–current
- Nathan Vaz, *Undergrad Research Mentor* 2026–current
- Matin Sadeghian, *Undergrad Research Mentor* 2026–current
- Steven Tang, *Undergrad Research Mentor* 2026–current
- Zaheen Jamil, *Undergrad Research Mentor* 2026–current
- Ailsa Sun, *Undergrad Research Mentor* 2026–current
- Kaustubh Khulbe, *Undergrad Research Mentor* .... 2025 spring
- Ishan Buyyanapragada, *ACM Mentor* 2024–2025
- Spencer Acquah, *ACM Mentor* 2023–2024
- Christopher Suarez, *ACM Mentor* 2023–2024

Service

- Berkeley Security Seminar cohost, ... 2026–current
- ACM mentor, ... 2022–2025
- UIUC course grader, ... 2023
- Self-organized competitive programming course instructor to 30 K-12 students, ... 2021
- Self-organized Java course instructor to over 100 K-12 students, ... 2020

Talks & Presentations

- Pixnapping: Bringing Pixel Stealing out of the Stone Age, *Berkeley Security Seminar* .... February 2025
- Pixnapping: Bringing Pixel Stealing out of the Stone Age, *Stanford Security Seminar* .... February 2025
- Pixnapping: Bringing Pixel Stealing out of the Stone Age, *SLICE Retreat* .... January 2025
- Pixnapping: Bringing Pixel Stealing out of the Stone Age, *ACM CCS* .... October 2025
- Peek-a-Walk: Leaking Secrets via Page Walk Side Channels, *Intel IPAS* .... July 2025
- Peek-a-Walk: Leaking Secrets via Page Walk Side Channels, *IEEE S&P (Oakland)* .... May 2025

Technical Skills

Programming Languages: C, C++, Python, x86 & ARM Assembly, OCaml, Bash, Rust, Java, Swift, SystemVerilog, Fortran, C#
Systems & Tools: Linux, ROS/ROS2, Git, LaTeX, LLVM, Android, iOS, macOS, Windows